Sizhe Wang

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Education

Washington University in St. Louis — **08 / 2025 - 05 / 2029 (Expected)**
PhD student in Computer Science, Advisor: Prof. Chenyang Lu — *St. Louis, U.S.*

University of Southern California — **01 / 2023 - 05 / 2025**
Master of Science in Computer Science — *Los Angeles, U.S.*

Sichuan University — **09 / 2018 - 06 / 2022**
Bachelor of Science in Physics — *Chengdu, China*

Research Experience

WashU | *LLM Post-Training & RL, Reliable AI, Multimodal Learning* — **08 / 2025 - Present**
Graduate Research Assistant, supervised by Prof. Chenyang Lu — *St. Louis, MO*

- Proposed Orthogonal Distillation for Preference Optimization (OrDiPO) for LLM post-training, pairing **Direct Preference Optimization (DPO)** with auxiliary teacher-guided knowledge distillation. Designed **preference-primary updates** that remove the distillation-gradient component parallel to an exponential moving average of past preference gradients while leaving the current DPO gradient unchanged, improving both preference accuracy and generation win rates over DPO and scalar-weighted joint training. [1]
- Exploring an online extension of OrDiPO's **gradient-projection approach** for LLM reasoning, combining Group Relative Policy Optimization (GRPO) with auxiliary **on-policy distillation (OPD)** on student-generated rollouts.
- Developed Clinical Uncertainty Risk Alignment (CURA) to reduce overconfidence in clinical language models. Built a two-stage pipeline combining task-specific fine-tuning with lightweight multi-head prediction, incorporating **individual-level uncertainty calibration** and **cohort-aware regularization** based on neighboring patients. Improved calibration across five MIMIC-IV tasks while maintaining competitive discrimination, with uncertainty-based triage analyses showing reduced overconfident false reassurance. [3]
- Introduced Router-Aware Specialist Team Optimization (RaSTO) for multimodal learning with missing modalities. Selected specialist teams by routed prediction loss and used response-space retrieval to estimate sample-specific competence and dynamically weight available specialists, improving discriminative performance across five clinical prediction tasks on TCGA and MIMIC. [2]

UNC | *LLM Alignment, Preference Optimization* — **06 / 2024 - 03 / 2025**
Research Assistant, supervised by Prof. Tianlong Chen — *Remote*

- Proposed Balanced Preference Optimization (BPO), a **clustering-based preference data selection method** for data-efficient LLM alignment. Combined embedding-based prompt selection, synthetic preference-pair generation, and clustering of projected LoRA gradients to adaptively allocate preference pairs across prompts. Improved Llama-3-8B's AlpacaEval win rate from 18.1% to 40.3% in the HH-RLHF setting, using 3.3K preference-training pairs versus 27K for vanilla DPO. [4]

Research Collaborations | *LLM Reasoning*

- Co-developed a **weakly supervised LLM reasoning** framework that combines supervised fine-tuning (SFT) with **iterative self-improvement**. Generated and filtered reasoning pairs from base and fine-tuned models on unlabeled questions, then applied preference learning to improve reasoning with limited human-annotated rationales. [6]
- Implemented **mistake tuning and self-rethinking prompting** for LLM reasoning, combining error-aware instruction tuning with inference-time refinement. Used explicitly labeled correct and incorrect Chain-of-Thought (CoT) rationales to guide training and iterative answer revision, improving performance across diverse reasoning tasks. [7]

Selected Publications

[1] **OrDiPO: Orthogonal Distillation for Preference Optimization in Language Models**
Under review, ICLR 2027 · **Sizhe Wang**, Daoyi Zhu, Peiqi Gao, Ziqi Xu, Jiezhong Xiao, Chenyang Lu

[2] **RaSTO: Router-Aware Specialist Team Optimization for Multimodal Clinical Risk Prediction**
Under review, ARR, Oct. 2026 · **Sizhe Wang**, Ziqi Xu, Liuyi Jin, Charles Alba, Peiqi Gao, Daoyi Zhu, Chenyang Lu

[3] **CURA: Clinical Uncertainty Risk Alignment for Language Model-Based Risk Prediction**
ACL 2026 (Main) · **Sizhe Wang**, Ziqi Xu, Claire Najjuuko, Charles Alba, Chenyang Lu

[4] **BPO: Towards Balanced Preference Optimization between Knowledge Breadth and Depth in Alignment**
NAACL 2025 (Main) · **Sizhe Wang**, Yongqi Tong, Hengyuan Zhang, Dawei Li, Xin Zhang, Tianlong Chen

[5] **ShifCon: Enhancing Non-Dominant Language Capabilities with a Shift-based Multilingual Contrastive Framework**
ACL 2025 (Main) · Hengyuan Zhang, Chenming Shang, **Sizhe Wang**, Dongdong Zhang, Yiyao Yu, Feng Yao, Renliang Sun, Yujiu Yang, Furu Wei

[6] **Optimizing Language Model's Reasoning Abilities with Weak Supervision**
arXiv preprint, 2024 · Yongqi Tong*, **Sizhe Wang***, Dawei Li, Yifan Wang, Simeng Han, Zi Lin, Chengsong Huang, Jiaxin Huang, Jingbo Shang

[7] **Can LLMs Learn from Previous Mistakes? Investigating LLMs' Errors to Boost for Reasoning**
ACL 2024 (Main) · Yongqi Tong, Dawei Li, **Sizhe Wang**, Yujia Wang, Fei Teng, Jingbo Shang

* Equal contribution.

Honors & Academic Service

Honors: Top 15% in the Periodic Review of Doctoral Students, WashU, 2026
Top Reviewer: NeurIPS 2026
Reviewer: ICLR 2027; ACL Rolling Review; IEEE/ACM TASLP; CIKM 2025

Technical Skills

Languages: Python, Java, C/C++, SQL, JavaScript/TypeScript, Swift
ML Frameworks & Tools: PyTorch, Hugging Face Transformers, vLLM, verl
Machine Learning: Large Language Model Post-Training, Knowledge Distillation, Multimodal Learning, Uncertainty Estimation
Development & Cloud: Docker, Kubernetes, Google Cloud Platform, Flask, Spring Boot, Node.js, SwiftUI